

Xufan Lu

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EDUCATION

Technical University of Munich , <i>M.Sc. Information Systems</i>	Apr 2024 – Oct 2026 (Expected) <i>Munich, Germany</i>
National University of Singapore , <i>Exchange Program in Computer Science</i>	Aug 2022 – Dec 2022 <i>Singapore</i>
Technical University of Munich , <i>B.Sc. Information Systems</i> <i>Top 2%, GPA 1.3 / 1.0</i>	Oct 2020 – Mar 2024 <i>Munich, Germany</i>

RESEARCH AND WORK EXPERIENCE

Research Aide <i>Argonne National Laboratory</i> Built Dr. XAS, an agentic AI assistant that enables non-coding domain experts to perform XAS data analysis within minutes instead of months of software training.	Jan 2026 – Apr 2026 <i>United States</i>
Data Scientist <i>Siemens AG, Analytics & AI Lab</i> - Delivered enterprise GenAI workflows on Azure AI Foundry, leveraging Azure-hosted model deployments, prompt flows, and managed compute for business-facing use cases. - Built a multi-agent system connected to Snowflake data to generate customer briefing reports. - Owned an internal Mendix OpenAI application for enterprise data, translating stakeholder needs into chat, summarization, and prompt-library features for up to 100 internal users.	Feb 2023 – Oct 2025 <i>Germany</i>
Volunteer Mentor & Coach <i>she.codes by TEC</i> Conducted Python workshops as a volunteer and mentored teenage girls to promote coding literacy.	Feb 2020 – Aug 2023 <i>Germany</i>
Student Tutor <i>Technical University of Munich</i> Delivered tutorials for Introduction to Information Systems.	Oct 2021 – Mar 2022 <i>Germany</i>

ACHIEVEMENTS

Max Weber-Programm Bayern Scholarship Awarded Bavaria's state scholarship for academic excellence, providing financial support and access to programs that foster personal development.	Apr 2022 – Sep 2026
1st Place HackaTUM 2024 <i>Wildtrack</i> Won 1st place by building ALEX, a multimodal lecture assistant, from concept to working prototype within a 48-hour team challenge.	Nov 2024
Best in TUM (Top 2% of Cohort) Recognized among the top 2% of students at the Technical University of Munich for outstanding academic performance.	Apr 2022 – Apr 2024

SELECTED PROJECTS

Dr. XAS — Agentic AI Assistant for X-ray Spectroscopy Analysis

Oct 2025 – Apr 2026

Tools: Python, TypeScript, AWS

- Designed an agentic analysis workflow for non-coding domain experts, integrating LLMs and explainable AI into X-ray spectroscopy workflows.
- Built and deployed the Dr. XAS analysis interface on Amazon Web Services, using Terraform to manage cloud infrastructure reproducibly.
- Reduced onboarding time for XAS data analysis from months to minutes while keeping outputs within expected accuracy ranges.

ALEX — Augmented Lecture Explainer (1st Place HackaTUM Wildtrack 2024)

Nov 2024

Tools: Python, LlamaIndex, Streamlit

- Built a multimodal Q&A assistant for live-streamed lectures, enabling students to query lecture content across audio transcripts and slides.
- Applied Whisper to transcribe lecture videos and synchronize timestamps with corresponding slide content.
- Used a self-hosted Llama 3.2 vision model to extract lecture slide content.
- Stored multimodal content in a vector database and applied RAG for grounded answers.

Bachelor's Thesis — Structured-Data Question-Answering Chatbot

Oct 2023 – Mar 2024

Tools: Python, LangChain, Azure, Streamlit

- Built a structured-data question-answering chatbot for mathematical and logical queries over tabular PDF data.
- Combined Azure Document Intelligence, Chroma vector search, and RAG to retrieve and reason over extracted PDF content.
- Implemented tool calling for on-demand calculations, improving reliability on quantitative questions beyond plain text generation.

Tick Talkers — Disease Spread Prediction Project

Nov 2022 – Jun 2024

Tools: Python, Statsmodels, Scikit-learn

- Built ARIMA-based time-series models to analyze environmental and human factors influencing the spread of Lyme Borreliosis.
- Completed as a 20-month TUM Junge Akademie scholarship project, covering research design, data collection, analysis, and presentation of findings.

Deep Learning Projects

Apr 2024 – Jun 2024

Tools: Python, PyTorch, PyTorch Lightning, TensorBoard, NumPy

- Implemented a CNN-based facial keypoint detector to predict 15 facial landmarks from image pixels, covering preprocessing, model training, and regression evaluation.
- Built CIFAR-10 image classification pipelines in PyTorch/PyTorch Lightning, applying data augmentation and TensorBoard monitoring to improve generalization across 10 object classes.

VolMe — Volunteer Event Platform

May 2024 – Aug 2024

Tools: Java, SQL, PostgreSQL, MongoDB, Docker

- Built a Dockerized volunteer-event platform using SQL/PostgreSQL and MongoDB to support event publishing, user selection, and participation workflows.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, JavaScript, TypeScript, C/C++, R

AI/ML Capabilities: ML/DL, computer vision, time-series forecasting, RAG, agentic AI, multi-agent systems, XAI

GenAI/Agent Frameworks: OpenAI Agents SDK, Microsoft Semantic Kernel, Microsoft Agent Framework, AutoGen, LangChain, LlamaIndex

Developer Tools: Git, Docker, Azure Cloud Platform, Amazon Web Services, TensorBoard

Libraries: Pandas, NumPy, PyTorch, TensorFlow, Keras, scikit-learn, React, Node.js, FastAPI